

Professional Summary:

Results-driven Senior Data Engineer with over 7+ years of progressive experience designing, building, and optimizing large-scale data pipelines, modern data architectures, and real-time analytics solutions. Proven expertise in AWS-centric cloud ecosystems, with strong contributions to Azure and GCP environments in high-impact sectors including healthcare, insurance, and retail.

Skilled at implementing robust data lakes, data warehouses, and streaming platforms to support advanced analytics, AI/ML pipelines, and enterprise BI needs. Adept at leading data migration, modernization, and cross-cloud integration initiatives, while ensuring security, governance, and cost-efficiency.

Recognized for engineering excellence in ETL/ELT automation, DevOps for data, and data quality assurance. Strong collaborator with cross-functional teams, with a track record of enhancing data reliability, accessibility, and performance at scale.

Technical Skills:

Big Data & ETL Frameworks:

Apache Spark (PySpark), Kafka, Hive, Pig, Sqoop, Flume, MapReduce, Delta Lake
ETL/ELT using Azure Data Factory, AWS Glue, Databricks, Apache Nifi

Cloud & Data Platforms:

AWS: S3, Redshift, Glue, EMR, Kinesis, Lake Formation, Lambda, Step Functions, Athena, CloudWatch, CodePipeline
Azure: Synapse, ADLS Gen2, Azure Data Factory, Event Hubs, Azure Databricks, Azure DevOps, Purview
GCP: BigQuery, Cloud Storage, cross-cloud integrations

Data Tools & Technologies:

Data Lakes, Data Warehousing, Real-time Streaming Pipelines
Power BI, Tableau, Amazon QuickSight (for visualization & analytics)
Apache Oozie, Airflow (workflow orchestration), Event-driven architectures

Security & Governance:

IAM, RBAC, Azure Purview, AWS Lake Formation, Kerberos, Data Encryption
HIPAA, PII data handling, compliance-focused pipeline design

Programming & Scripting:

Python, SQL (T-SQL, HiveQL), Shell scripting
Terraform, CloudFormation (IaC for data pipelines)

DevOps & CI/CD:

Azure DevOps, AWS CodePipeline, GitHub Actions
Docker (for testing data components), **Monitoring via CloudWatch & Alerts**

Data Integration & APIs:

RESTful APIs, JSON/XML processing, 3rd-party API ingestion
Kafka Connect, Custom Connectors (Azure ↔ AWS ↔ GCP)

Professional Experience

Senior Data Engineer – Kaiser Permanente – Oakland, CA | August 2024 – Present

- Architected scalable and fault-tolerant data pipelines using **AWS Glue**, **Lambda**, and **Step Functions** to automate data ingestion, transformation, and validation across large healthcare data sets.
- Designed and implemented **Amazon Redshift** data warehouse solutions optimized with sort keys, distribution styles, and materialized views, resulting in a 60% reduction in query latency for enterprise reporting.
- Leveraged **Amazon S3**, **Athena**, and **Lake Formation** to build a secure, centralized data lake supporting compliance with HIPAA and internal governance standards for patient data.
- Built and managed **real-time streaming pipelines** using **Amazon Kinesis** and **Apache Kafka** on **AWS MSK**, enabling event-driven analytics for patient monitoring systems and operational alerts.
- Developed ETL jobs using **AWS Glue Studio** with **PySpark**, enabling processing of complex hierarchical and nested healthcare data formats (HL7, JSON, XML) at scale.
- Automated end-to-end deployment and monitoring of data workflows using **Terraform**, **CloudFormation**, and **AWS CloudWatch**, enabling Infrastructure as Code (IaC) practices and system observability.

- Integrated third-party data feeds via **AWS API Gateway** and **Lambda**, transforming and enriching provider and clinical data for use in downstream analytics.
- Implemented granular data access policies using **AWS Lake Formation** and **IAM roles**, ensuring fine-grained security, auditability, and compliance across multi-team analytics platforms.
- Led the migration of legacy on-prem Hadoop workloads to **AWS EMR**, optimizing job performance through Spark tuning, autoscaling, and dynamic allocation strategies.
- Developed reusable libraries and internal packages in **Python** and **PySpark** to accelerate development for ingestion, transformation, and validation across multiple AWS services.
- Collaborated on cross-cloud architecture, integrating **Azure Data Lake Storage (Gen2)** with **AWS Glue** via custom connectors to unify analytics from multiple business units.
- Designed multi-cloud interoperability solutions by enabling **GCP BigQuery** and **AWS Redshift** integration through **data federation and cross-cloud replication**, supporting enterprise data sharing initiatives.
- Led knowledge-sharing sessions and mentoring programs on **data engineering best practices**, **AWS cost optimization**, and **security-first development**, enhancing team productivity and cloud maturity.
- Partnered with DevOps teams to build secure and resilient **CI/CD pipelines for data workloads** using **AWS CodePipeline**, CodeBuild, and integration with Azure DevOps for hybrid deployment strategies.

Environment: AWS (Glue, Lambda, Step Functions, S3, Redshift, Athena, Lake Formation, Kinesis, MSK, EMR, IAM, CloudWatch), Apache Spark (PySpark), Kafka, Hadoop, Python, SQL, Terraform, CloudFormation, CI/CD, Azure Data Lake Gen2, GCP BigQuery, HIPAA-compliant multi-cloud data architecture.

Data Engineer – Nationwide Mutual Insurance – Columbus, OH | June 2021 – July 2024

- Designed and maintained scalable data pipelines using **Azure Data Factory (ADF)** to orchestrate ETL workflows across structured, semi-structured, and unstructured data sources, enhancing data refresh reliability and automation.
- Engineered and optimized data lake architecture on **Azure Data Lake Storage Gen2**, establishing robust folder hierarchies, lifecycle policies, and data zone partitioning to support advanced analytics use cases.
- Developed and deployed **Azure Synapse Analytics** solutions for enterprise-wide reporting and big data processing, integrating Spark pools and SQL on-demand queries for cost-effective ad hoc exploration.
- Implemented real-time and batch ingestion pipelines using **Azure Event Hubs**, **Kafka**, and **Stream Analytics**, enabling near-real-time insights for risk modeling and customer behavior analytics.
- Collaborated with cross-functional teams to design and implement **Delta Lake architecture** on Azure Databricks, supporting ACID transactions and unified batch/stream processing pipelines.
- Developed end-to-end **PySpark** and **SQL-based** data transformation pipelines on **Azure Databricks**, reducing data processing times for actuarial models by over 55%.
- Managed data integration with **external APIs**, on-prem SQL Server, Oracle, and RESTful sources, and ingested the data into Azure Blob Storage for centralized access and downstream analytics.
- Created and maintained **CI/CD pipelines** for data projects using **Azure DevOps**, ensuring version control, test automation, and seamless production deployments.
- Built dynamic dashboards and operational monitoring tools using **Power BI** integrated with Azure Data Services, providing real-time visibility to data quality and pipeline health.
- Implemented robust data governance and security using **Azure Purview**, **Key Vault**, and **RBAC**, ensuring compliance with enterprise policies, PII data protection, and role-based access controls.
- Designed and migrated legacy ETL jobs to the cloud using **Azure Data Factory**, replacing SSIS packages and reducing operational overhead while improving pipeline reliability.
- Integrated **AWS S3** and **Google Cloud Storage (GCS)** data feeds into the Azure ecosystem using **ADF connectors** and **Databricks multi-cloud capabilities**, enabling centralized analytics from multi-cloud data sources.
- Tuned and scaled **Databricks Spark clusters** for performance optimization, memory management, and resource utilization, improving large-scale data model training and downstream data processing.
- Participated in **data mesh** implementation initiatives, building domain-driven data products and exposing them via APIs and Azure Data Catalog, accelerating discoverability and self-service analytics across business units.

Environment: Azure (Data Factory, Data Lake Storage Gen2, Synapse Analytics, Event Hubs, Stream Analytics, Databricks, Blob Storage, Purview, Key Vault, RBAC), Apache Spark (PySpark), Delta Lake, Kafka, SQL, Python, Power BI, Azure DevOps CI/CD, REST APIs, SQL Server, Oracle, multi-cloud integration (AWS S3, GCS), Data Mesh architecture.

Big Data Developer – Home Depot – Atlanta, GA | September 2018 – April 2021

- Designed and developed robust ETL pipelines using **Apache Sqoop**, **Apache Flume**, and **Apache Nifi** to ingest large-scale structured and unstructured data from various RDBMS, logs, and streaming sources into HDFS for downstream analytics.
- Utilized **Apache Hadoop** ecosystem (HDFS, MapReduce, Hive, Pig) to process and analyze multi-terabyte datasets efficiently, improving batch data processing performance by over 40% compared to traditional systems.
- Developed complex data transformation logic using **Apache Pig** and **HiveQL**, enabling business users to access actionable insights from raw clickstream and transactional data.
- Built reusable MapReduce jobs to perform data cleansing, validation, and enrichment for retail sales and inventory datasets, reducing manual data wrangling effort by 60%.
- Implemented data warehousing solutions using **Apache Hive** on Hadoop, optimizing partitioning, bucketing, and indexing strategies to significantly enhance query performance and reporting latency.
- Created and maintained large-scale, scheduled data workflows using **Apache Oozie**, ensuring timely execution of dependent jobs and alerting mechanisms for failure scenarios.
- Designed real-time log ingestion frameworks using **Apache Flume**, enabling proactive monitoring of application behavior and system performance for in-store retail applications.
- Engineered efficient **data archival** and retrieval systems leveraging **HDFS**, applying compression techniques such as Snappy and Avro to reduce storage footprint by over 30%.
- Developed data parsing and normalization scripts using **Python** and **Shell scripting**, automating routine data preparation tasks across multiple data feeds and formats (CSV, XML, JSON).
- Coordinated with data science and BI teams to deliver clean and reliable data sets from the Hadoop platform, ensuring consistency and integrity in ML model training and dashboard visualizations.
- Tuned and configured **YARN**, **HDFS**, and **MapReduce** job parameters for better resource utilization and cluster performance, reducing job failure rates and optimizing compute cost.
- Participated in migration efforts from legacy SQL-based reporting systems to Hadoop-based distributed storage and computation, improving scalability and processing speed across teams.
- Actively contributed to setting up **data governance**, security policies, and access controls on Hadoop systems using tools like **Apache Ranger** and **Kerberos**, ensuring compliance with enterprise standards and data privacy regulations.

Environment: Apache Hadoop ecosystem (HDFS, YARN, MapReduce, Hive, Pig), Apache Sqoop, Flume, NiFi, Oozie, HiveQL, Python, Shell scripting, Avro, Snappy, Kerberos, Apache Ranger, RDBMS sources, distributed batch and real-time data processing.

Education: Bachelors in Business Analytics, University of North Texas, Denton, Texas